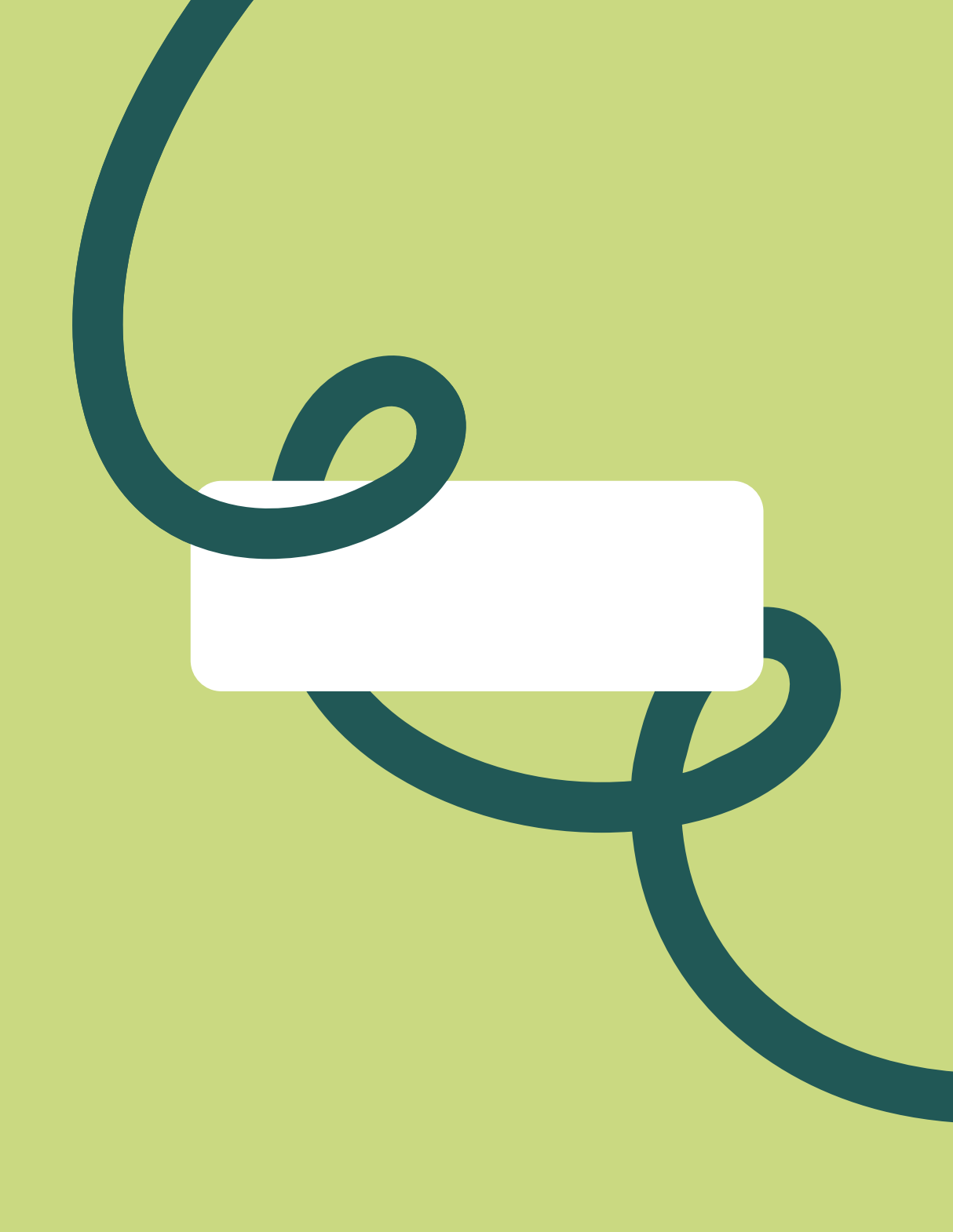
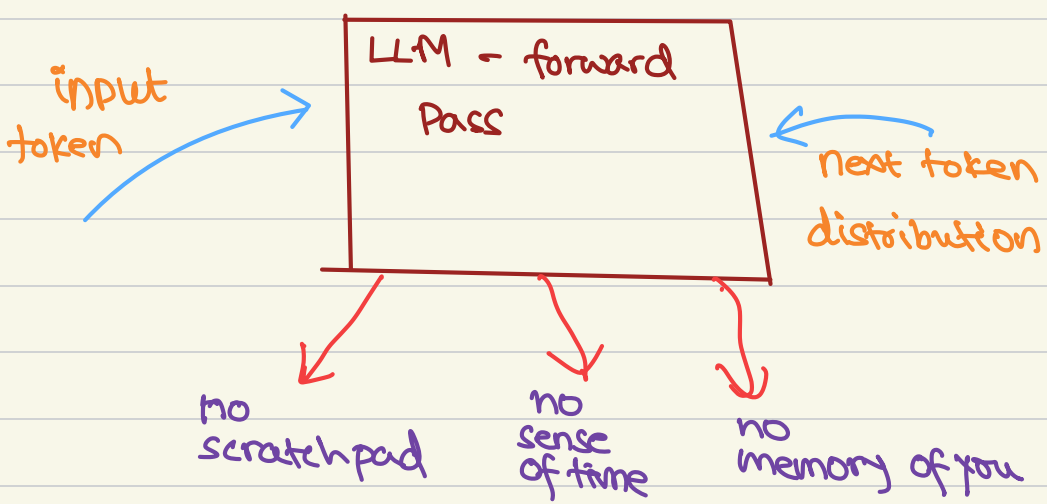


CHAPTER 1 – The Stateless problem

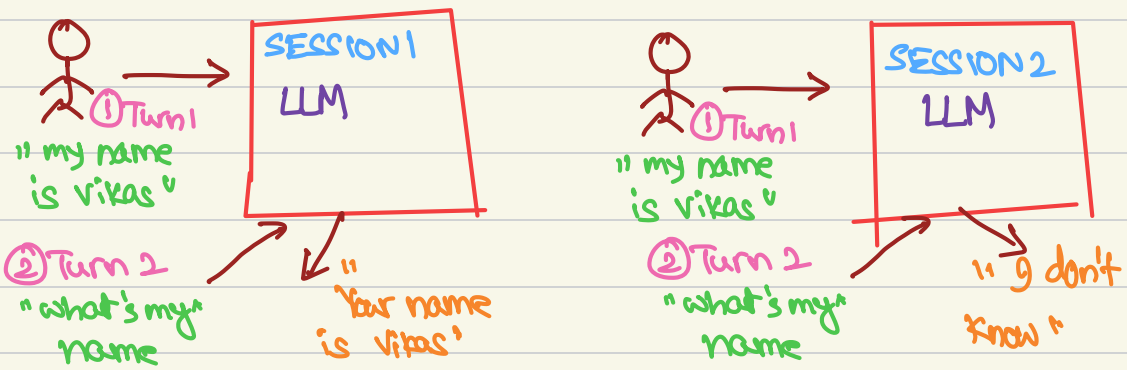
- Memory in AI system is stateless. It means that model doesn't remember anything but an illusion of model seems to remember comes.
- Any of the weights are not modified during any of the interaction with LLM



A raw LLM is a stateless pure function. It doesn't remember anything after it returns.

ILLUSION OF MEMORY

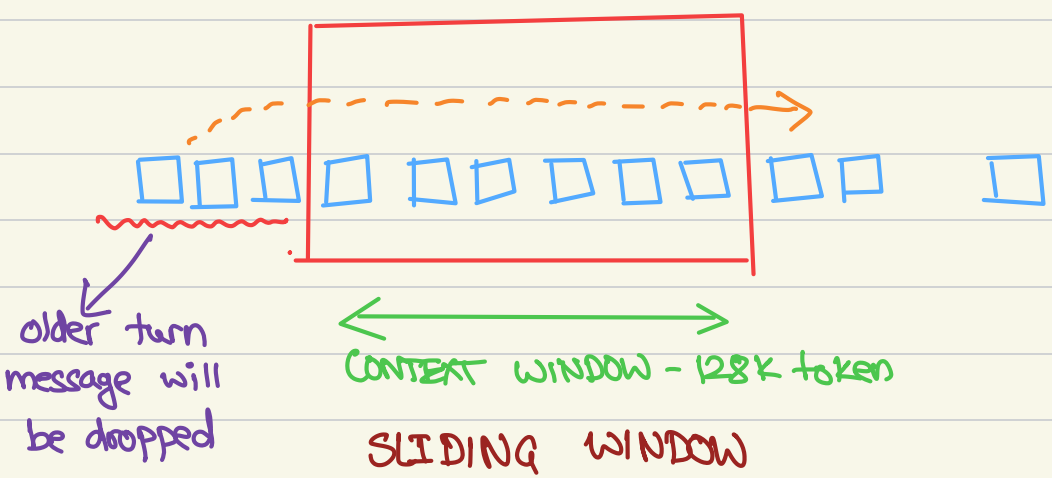
- We may get a feel of that AI model remembers thing. For example, having conversation with AI in one session starting and stating your name and asking name after 3-4 turn, you may get your name back. However this seems to be an illusion as model doesn't remember anything (in general). What actually happens is that in every conversation whole message, including the previous message is sent back to model.
- In a pure native LLM, we can test this by creating another session and asking about name, it will not have any clue.



CONSEQUENCES

- There are 2 consequenses
  (i) Every replay doesn't scale
  (ii) Reaching context window limit.

- **Replay** - Sending everything in all LLM calls.
  e.g. token usage 50 → 100 → 250 → 350 .....
- **Context window** - Every LLM will have context window budget. e.g. 128K token. Once it reaches the limit older turns needs to be dropped.



"JUST MAKE CONTEXT WINDOW BIGGER"

- This will not solve the problem; rather delay the problem.
  - In every turn the whole message is sent which makes this approach quadratically costly. e.g. 50 → 150 → 250 → 350 token.
  - Another problem is that sending too much information to LLM dilutes the purpose and problem of "Lost in the middle" comes.
  - It means, for example of 128 K token window if only 3 messages are important then whole attention of LLM is diluted.

